

KAIROS - Previous vs New Comparison

Simple summary of benchmark quality and SIH problem-statement alignment

Metric	Previous comparison	New fair comparison	Improvement
Task	LR classified the current window; GNN predicted a future window.	LR and KAIROS both predict the same 60-second future window.	Now apples-to-apples
F1	Old GNN: 0.3545 Old LR: 0.7097	KAIROS: 0.7390 LR: 0.6770	+0.0620
Precision	Old GNN: 0.5027 Old LR: 0.6535	KAIROS: 0.6696 LR: 0.6331	+0.0365
Recall	Old GNN: 0.2738 Old LR: 0.7765	KAIROS: 0.8245 LR: 0.7275	+0.0971
False-positive rate	Old GNN: 0.1694 Old LR: 0.2578	KAIROS: 0.2548 LR: 0.2640	-0.0092 (better)
ROC-AUC	Old GNN: 0.5813 Old LR: not reported	KAIROS: 0.8604 LR: 0.7923	+0.0681

Project area	Previous	New	Current status
Benchmark validity	Different prediction tasks were compared.	Same features, target, calibration, and untouched test windows.	Fixed
CTU-13 validation	Scenario 6 development smoke test only.	Train S6, validate S11, untouched zero-shot test on S12.	Implemented; FPR remains high
PCAP support	Mostly backend tooling / CSV-first flow.	CSV, PCAP, and PCAPNG work through the upload pipeline.	Implemented
MITRE stages	Six outputs could appear despite limited training support.	Unsupported classes are masked; coverage is shown explicitly.	Partial: 3 of 6 stages supervised
Frontend	Functional but limited.	Functional ingestion and forecast display.	Needs visual polish

Final verdict: The new version is substantially better aligned with the SIH problem statement because its main performance claim now uses a fair future-forecasting benchmark. The most important remaining technical issue is the high false-positive rate on unseen CTU-13 Scenario 12; the main presentation issue is frontend polish.